# STT 843: Multivariate Analysis

## 9. Canonical Variates and Canonical Correlation (Chapter 10.1-10.4

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# Outline

1 Canonical Variates and Canonical Correlation

- Canonical Correlatin analysis focuses on the correlation between a linear combination of the variable in one set and a linear combination of the variables in another set.
- Canonical variables, Canonical correlation.
- Examples: Arithmetic speed and arithmetic power to reading speed and reading power, Governmental policy variables with economic goal variables, College "performance" variables with precollege "achievement" variables.

# Canonical Variates and Canonical Correlation

Let $\boldsymbol{X}^{(1)}$ be a $p \times 1$ random vector, $\boldsymbol{X}^{(2)}$ be a $q \times 1$ random vector with $p \leq q$, and

$$\mathrm{E}\left(\boldsymbol{X}^{(1)}\right) = \boldsymbol{\mu}^{(1)}; \quad \mathrm{Cov}\left(\boldsymbol{X}^{(1)}\right) = \Sigma_{11};$$
$$\mathrm{E}\left(\boldsymbol{X}^{(2)}\right) = \boldsymbol{\mu}^{(2)}; \quad \mathrm{Cov}\left(\boldsymbol{X}^{(2)}\right) = \Sigma_{22};$$
$$\mathrm{Cov}\left(\boldsymbol{X}^{(1)}, \boldsymbol{X}^{(2)}\right) = \Sigma_{12} = \Sigma_{21}^{\mathsf{T}}.$$

Set

$$U = \mathbf{a}^{\mathsf{T}}\boldsymbol{X}^{(1)}, \quad V = \mathbf{b}^{\mathsf{T}}\boldsymbol{X}^{(2)},$$

Then we shall seek coefficient vectors $\mathbf{a}$ and $\mathbf{b}$ such that

$$\mathrm{Corr}(U, V) = \frac{\mathbf{a}^\top \Sigma_{12} \mathbf{b}}{\sqrt{\mathbf{a}^\top \Sigma_{11} \mathbf{a}} \sqrt{\mathbf{b}^\top \Sigma_{22} \mathbf{b}}}$$

is as large as possible.

- The first pair of canonical variables, or first canonical variate pair is the pair of linear combination $U_1$ and $V_1$ having unit variances, which maximize the correlation $\mathrm{Corr}(U, V)$;

- The second pair of canonical variables, or second canonical variate pair
  the pair of linear combinations $U_2$ and $V_2$ having unit variances, which maximize the correlation $\mathrm{Corr}(U, V)$ among all choices that are uncorrelated with the first pair of canonical variables.
- The $k$th pair of canonical variables, or $k$ th canonical variate pair
  the pair of linear combinations $U_k, V_k$ having unit variances, which maximize the correlation $\mathrm{Corr}(U, V)$ among all choices uncorrelated with the previous $k-1$ canonical variable pares

The correlation between the $k$th pair of canonical variable $s$ is called the $k$th canonical correlation

# Result 9-1.1

Suppose $p \leq q$ and let the $p$-dimensional random vectors $\boldsymbol{X}^{(1)}$ and $q$ dimensional $\boldsymbol{X}^{(2)}$ have $\mathrm{Cov}\left(\boldsymbol{X}^{(1)}\right) = \Sigma_{11}, \mathrm{Cov}\left(\boldsymbol{X}^{(1)}\right) = \Sigma_{22}$, and $\mathrm{Cov}\left(\boldsymbol{X}^{(1)}, \boldsymbol{X}^{(2)}\right) = \Sigma_{12}$, where $\Sigma$ has full rank. For coefficients $p \times 1$ vector a and $q \times 1$ vector $\mathbf{b}$, form the linear combination $U = \mathbf{a}^{\mathsf{T}} \boldsymbol{X}^{(1)}$ and $V = \mathbf{b}^{\mathsf{T}} \boldsymbol{X}^{(2)}$. Then

$$\max_{\mathbf{a},\mathbf{b}} \mathrm{Corr}(U, V) = \rho_1^*$$

attained by the linear combinations (first canonical variate pair)

$$U_1 = \mathbf{e}_1^\top \Sigma_{11}^{-1/2} \boldsymbol{X}^{(1)} \quad \text{and} \quad V_1 = \mathbf{f}_1^\top \Sigma_{22}^{-1/2} \boldsymbol{X}^{(2)}$$

The $k$th pair of canonical variates, $k = 2, 3, \ldots, p$

$$U_k = \mathbf{e}_k^\top \Sigma_{11}^{-1/2} \boldsymbol{X}^{(1)} \quad \text{and} \quad V_k = \mathbf{f}_k^\top \Sigma_{22}^{-1/2} \boldsymbol{X}^{(2)}$$

maximize

$$\max_{\mathbf{a},\mathbf{b}} \mathrm{Corr}(U, V) = \rho_k^*$$

among those linear combinations uncorrelated with the preceding $1, 2, \ldots, k-1$ canonical variables.

- Here $\rho_1^{*2} \geq \rho_2^{*2} \cdots \geq \rho_p^{*2}$ are the eigenvalues of $\Sigma_{11}^{-1/2}\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\Sigma_{11}^{-1/2}$, and $\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_p$ are associated $p \times 1$ eigenvectors.
- The quantities $\rho_1^{*2} \geq \rho_2^{*2} \cdots \geq \rho_p^{*2}$ are also the $p$ largest eigenvalues of the matrix $\Sigma_{22}^{-1/2}\Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22}^{-1/2}$ with corresponding $q \times 1$ eigenvectors $\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_p$.
- Each $\mathbf{f}_i$ is proportional to $\Sigma_{22}^{-1/2}\Sigma_{21}\Sigma_{11}^{-1/2}\mathbf{e}_i$.

- The canonical variates have the properties

$$\mathrm{Var}\left(U_k\right) = \mathrm{Var}\left(V_k\right) = 1, k = 1, \ldots, p,$$
$$\mathrm{Corr}\left(U_k, U_\ell\right) = \mathrm{Corr}\left(V_k, V_\ell\right) = \mathrm{Corr}\left(U_k, V_\ell\right) = 0$$

for $k, \ell = 1, 2, \ldots, p$ and $k \neq \ell$.

# Example 9-1.1

Suppose $\mathbf{Z}^{(1)} = \left[Z_1^{(1)}, Z_2^{(1)}\right]^{\mathsf{T}}$ are standardize variables and $\mathbf{Z}^{(2)} = \left[Z_1^{(2)}, Z_2^{(2)}\right]^{\mathsf{T}}$ are also standardized variables. Let $\mathbf{Z} = \left[\mathbf{Z}^{(1)}, \mathbf{Z}^{(2)}\right]^{\mathsf{T}}$ and

$$\mathrm{Cov}(\mathbf{Z}) = \begin{bmatrix} 1.0 & 4 & .5 & .6 \\ .4 & 1.0 & .3 & .4 \\ .5 & .3 & 1.0 & .2 \\ .6 & .4 & .2 & 1.0 \end{bmatrix}$$

Calculate canonical variates and canonical correlations for standardized variables $\mathbf{Z}^{(1)}$ and $\mathbf{Z}^{(2)}$.

# Example 9-1.2

Compute the computing correlations between the first pair canonical variates and their component variables for the situation considered in Example 9-1.1.

# Example 9-1.3

Consider the covariance matrix

$$\mathrm{Cov}\begin{pmatrix} X_1^{(1)} \\ X_2^{(1)} \\ X_1^{(2)} \\ X_2^{(2)} \end{pmatrix} = \begin{bmatrix} 100 & 0 & 0 & 0 \\ 0 & 1 & 0.95 & 0 \\ 0 & 0.95 & 1 & 0 \\ 0 & 0 & 0 & 100 \end{bmatrix}$$

Calculate the canonical correlation between $\left[X_1^{(1)}, X_2^{(1)}\right]^\mathsf{T}$ and $\left[X_1^{(2)}, X_2^{(2)}\right]^\mathsf{T}$

# The Sample Canonical Variates and Sample Canonical Correlation

Results 9-1.2. Let $\hat{\rho}_1^{*2} \geq \hat{\rho}_2^{*2} \geq \ldots \geq \hat{\rho}_p^{*2}$ be the $p$ ordered eigenvalues of $S_{11}^{-1/2} S_{12} S_{22}^{-1} S_{21} S_{11}^{-1/2}$ with corresponding eigenvectors $\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \ldots, \hat{\mathbf{e}}_p$, where $p \leq q$.
Let $\hat{\mathbf{f}}_1, \ldots, \hat{\mathbf{f}}_p$ be the eigenvectors of $S_{22}^{-1/2} S_{21} S_{11}^{-1} S_{12} S_{22}^{-1/2}$. Then the $k$ th sample canonical variate pair is

$$\hat{U}_k = \hat{\mathbf{e}}_k S_{11}^{-1/2} \mathbf{x}^{(1)}, \quad \hat{V}_k = \hat{\mathbf{f}}_k S_{22}^{-1/2} \mathbf{x}^{(2)}$$

where $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$ are the values of the variables $\boldsymbol{X}^{(1)}$ and $\boldsymbol{X}^{(2)}$ for a particular experimental unit. Also for the $k$ th pair, $k = 1, \ldots, p$.

$$r_{\hat{U}_k, \hat{V}_k} = \hat{\rho}_k^*.$$

The quantities $\hat{\rho}_1^*, \ldots, \hat{\rho}_p^*$ are the sample canonical correlations.

We define

$$\begin{aligned}
\mathbf{R}_{\hat{\mathbf{U}},\mathbf{x}^{(1)}} &= \text{ matrix of sample correlations of } \hat{\mathbf{U}} \text{ with } \mathbf{x}^{(1)} \\
\mathbf{R}_{\hat{\mathbf{V}},\mathbf{x}^{(2)}} &= \text{ matrix of sample correlations of } \hat{\mathbf{V}} \text{ with } \mathbf{x}^{(2)} \\
\mathbf{R}_{\hat{\mathbf{U}},\mathbf{x}^{(2)}} &= \text{ matrix of sample correlations of } \hat{\mathbf{U}} \text{ with } \mathbf{x}^{(2)} \\
\mathbf{R}_{\hat{\mathbf{V}},\mathbf{x}^{(1)}} &= \text{ matrix of sample correlations of } \hat{\mathbf{V}} \text{ with } \mathbf{x}^{(1)}
\end{aligned}$$

Corresponding to (10-19), we have

$$\mathbf{R}_{\hat{\mathbf{U}},\mathbf{x}^{(1)}} = \hat{\mathbf{A}}\mathbf{S}_{11}\mathbf{D}_{11}^{-1/2}$$
$$\mathbf{R}_{\hat{\mathbf{V}},\mathbf{x}^{(2)}} = \hat{\mathbf{B}}\mathbf{S}_{22}\mathbf{D}_{22}^{-1/2}$$
$$\mathbf{R}_{\hat{\mathbf{U}},\mathbf{x}^{(2)}} = \hat{\mathbf{A}}\mathbf{S}_{12}\mathbf{D}_{22}^{-1/2}$$
$$\mathbf{R}_{\hat{\mathbf{V}},\mathbf{x}^{(1)}} = \hat{\mathbf{B}}\mathbf{S}_{21}\mathbf{D}_{11}^{-1/2}$$

where $\mathbf{D}_{11}^{-1/2}$ is the $(p \times p)$ diagonal matrix with $i$ th diagonal element $\left(\text{sample var}\left(x_i^{(1)}\right)\right)^{-1/2}$ and $\mathbf{D}_{22}^{-1/2}$ is the $(q \times q)$ diagonal matrix with $i$th diagonal element $\left(\text{ sample var}\left(x_i^{(2)}\right)\right)^{-1/2}$.

# Comment

If the observations are standardized, the data matrix becomes

$$\mathbf{Z} = \begin{bmatrix} \mathbf{Z}^{(1)} & \mathbf{Z}^{(2)} \end{bmatrix} = \begin{bmatrix} \mathbf{Z}_1^{(1)\mathsf{T}} & \mathbf{Z}_1^{(2)\mathsf{T}} \\ \vdots & \vdots \\ \mathbf{Z}_n^{(1)\mathsf{T}} & \mathbf{Z}_n^{(2)\mathsf{T}} \end{bmatrix}$$

and the sample canonical variates become

$$\underset{(p\times 1)}{\hat{\mathbf{U}}} = \hat{\mathbf{A}}_{\mathbf{z}}\mathbf{z}^{(1)}, \quad \underset{(q\times 1)}{\hat{\mathbf{V}}} = \hat{\mathbf{B}}_{\mathbf{z}}\mathbf{z}^{(2)},$$

where $\hat{\mathbf{A}}_{\mathbf{z}} = \hat{\mathbf{A}}\mathbf{D}_{11}^{1/2}$ and $\hat{\mathbf{B}}_{\mathbf{z}} = \hat{\mathbf{B}}\mathbf{D}_{22}^{1/2}$.

- The sample canonical correlations are unaffected by the standardization.
- The correlations displayed in (10-29) remain unchanged and may be calculated, for standardized observations, by substituting $\hat{\mathbf{A}}_{\mathbf{z}}$ for $\hat{\mathbf{A}}$, $\hat{\mathbf{B}}_{\mathbf{z}}$ for $\hat{\mathbf{B}}$, and $\mathbf{R}$ for $\mathbf{S}$.
- Note that $\mathbf{D}_{11}^{-1/2} = \underset{(p\times p)}{\mathbf{I}}$ and $\mathbf{D}_{22}^{-1/2} = \underset{(q\times q)}{\mathbf{I}}$ for standardized observations.

# Example 9.4 (Canonical correlation analysis of the chicken-bone data)

In Example 9.4, data consisting of bone and skull measurements of white leghorn fowl were described. From this example, the chicken-bone measurements for

$$\text{Head}\left(\mathbf{X}^{(1)}\right): \begin{cases} X_1^{(1)} = \text{skull length} \\ X_2^{(1)} = \text{skull breadth} \end{cases}$$

$$\text{Leg}\left(\mathbf{X}^{(2)}\right): \begin{cases} X_1^{(2)} = \text{femur length} \\ X_2^{(2)} = \text{tibia length} \end{cases}$$

have the sample correlation matrix

$$\mathbf{R} = \begin{bmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{R}_{21} & \mathbf{R}_{22} \end{bmatrix} = \begin{bmatrix} 1.0 & .505 & .569 & .602 \\ .505 & 1.0 & .422 & .467 \\ .569 & .422 & 1.0 & .926 \\ .602 & .467 & .926 & 1.0 \end{bmatrix}$$

A canonical correlation analysis of the head and leg sets of variables using **R** produces the two canonical correlations and corresponding pairs of variables

$$\widehat{\rho_1^*} = .631$$

$$\hat{U}_1 = .781 z_1^{(1)} + .345 z_2^{(1)}$$

$$\hat{V}_1 = .060 z_1^{(2)} + .944 z_2^{(2)}$$

and

$$\widehat{\rho_2^*} = .057$$

$$\hat{U}_2 = -.856 z_1^{(1)} + 1.106 z_2^{(1)}$$

$$\hat{V}_2 = -2.648 z_1^{(2)} + 2.475 z_2^{(2)}$$

Here $z_i^{(1)}, i = 1, 2$ and $z_i^{(2)}, i = 1, 2$ are the standardized data values for sets 1 and 2 , respectively.

# Example 10.5 (Canonical correlation analysis of job satisfaction)

As part of a larger study of the effects of organizational structure on "job satisfaction," Dunham [4] investigated the extent to which measures of job satisfaction are related to job characteristics. Using a survey instrument, Dunham obtained measurements of $p = 5$ job characteristics and $q = 7$ job satisfaction variables for $n = 784$ executives from the corporate branch of a large retail merchandising corporation. Are measures of job satisfaction associated with job characteristics? The answer may have implications for job design.

The original job characteristic variables, $\mathbf{X}^{(1)}$, and job satisfaction variables, $\mathbf{X}^{(2)}$, were respectively defined as

$$\mathbf{X}^{(1)} = \begin{bmatrix} X_1^{(1)} \\ X_2^{(1)} \\ X_3^{(1)} \\ X_4^{(1)} \\ X_5^{(1)} \end{bmatrix} = \begin{bmatrix} \text{feedback} \\ \text{task significance} \\ \text{task variety} \\ \text{task identity} \\ \text{autonomy} \end{bmatrix}$$

$$\mathbf{X}^{(2)} = \begin{bmatrix} X_1^{(2)} \\ X_2^{(2)} \\ X_3^{(2)} \\ X_4^{(2)} \\ X_5^{(2)} \\ X_6^{(2)} \\ X_7^{(2)} \end{bmatrix} = \begin{bmatrix} \text{supervisor satisfaction} \\ \text{career-future satisfaction} \\ \text{financial satisfaction} \\ \text{workload satisfaction} \\ \text{company identification} \\ \text{kind-of-work-satisfaction} \\ \text{general satisfaction} \end{bmatrix}$$

Responses for variables $\mathbf{X}^{(1)}$ and $\mathbf{X}^{(2)}$ were recorded on a scale and then standardized. The sample correlation matrix based on 784 responses is

$$\mathbf{R} = \begin{bmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{R}_{21} & \mathbf{R}_{22} \end{bmatrix}$$

$$= \begin{bmatrix} 1.0 & & & & & .33 & .32 & .20 & .19 & .30 & .37 & .21 \\ .49 & 1.0 & & & & .30 & .21 & .16 & .08 & .27 & .35 & .20 \\ .53 & .57 & 1.0 & & & .31 & .23 & .14 & .07 & .24 & .37 & .18 \\ .49 & .46 & .48 & 1.0 & & .24 & .22 & .12 & .19 & .21 & .29 & .16 \\ .51 & .53 & .57 & .57 & 1.0 & .38 & .32 & .17 & .23 & .32 & .36 & .27 \\ .33 & .30 & .31 & .24 & .38 & 1.0 & & & & & & \\ .32 & .21 & .23 & .22 & .32 & .43 & 1.0 & & & & & \\ .20 & .16 & .14 & .12 & .17 & .27 & .33 & 1.0 & & & & \\ .19 & .08 & .07 & .19 & .23 & .24 & .26 & .25 & 1.0 & & & \\ .30 & .27 & .24 & .21 & .32 & .34 & .54 & .46 & .28 & 1.0 & & \\ .37 & .35 & .37 & .29 & .36 & .37 & .32 & .29 & .30 & .35 & 1.0 & \\ .21 & .20 & .18 & .16 & .27 & .40 & .58 & .45 & .27 & .59 & .31 & 1.0 \end{bmatrix}$$

The $\min(p, q) = \min(5, 7) = 5$ sample canonical correlations and the sample canonical variate coefficient vectors (from Dunham [4]) are displayed in the table.
For example, the first sample canonical variate pair is

$$\hat{U}_1 = .42z_1^{(1)} + .21z_2^{(1)} + .17z_3^{(1)} - .02z_4^{(1)} + .44z_5^{(1)}$$
$$\hat{V}_1 = .42z_1^{(2)} + .22z_2^{(2)} - .03z_3^{(2)} + .01z_4^{(2)} + .29z_5^{(2)} + .52z_6^{(2)} - .12z_7^{(2)}$$

with sample canonical correlation $\widehat{\rho_1^*} = .55$.

- According to the coefficients, $\hat{U}_1$ is primarily a feedback and autonomy variable, while $\hat{V}_1$ represents supervisor, career-future, and kind-of-work satisfaction, along with company identification.
- To provide interpretations for $\hat{U}_1$ and $\hat{V}_1$, the sample correlations between $\hat{U}_1$ and its component variables and between $\hat{V}_1$ and its component variables were computed. Also, the following table shows the sample correlations between variables in one set and the first sample canonical variate of the other set.

# Sample Correlations Between Original Variables and Canonical Variables

| | Sample canonical variates | | | Sample canonical variates | |
|---|---|---|---|---|---|
| $\mathbf{X}^{(1)}$ variables | $\hat{U}_1$ | $\hat{V}_1$ | $\mathbf{X}^{(2)}$ variables | $\hat{U}_1$ | $\hat{V}_1$ |
| 1. Feedback | .83 | .46 | 1. Supervisor satisfaction | .42 | .75 |
| 2. Task significance | .74 | .41 | 2. Career-future satisfaction | .35 | .65 |
| 3. Task variety | .75 | .42 | 3. Financial satisfaction | .21 | .39 |
| 4. Task identity | .62 | .34 | 4. Workload satisfaction | .21 | .37 |
| 5. Autonomy | .85 | .48 | 5. Company identification | .36 | .65 |
| | | | 6. Kind-of-work satisfaction | .44 | .80 |
| | | | 7. General satisfaction | .28 | .50 |

All five job characteristic variables have roughly the same correlations with the first canonical variate $\hat{U}_1$. From this standpoint, $\hat{U}_1$ might be interpreted as a job characteristic "index." This differs from the preferred interpretation, based on coefficients, where the task variables are not important.

- The other member of the first canonical variate pair, $\hat{V}_1$, seems to be representing, primarily, supervisor satisfaction, career-future satisfaction, company identification, and kind-of-work satisfaction. As the variables suggest, $\hat{V}_1$ might be regarded as a job satisfaction-company identification index.
- This agrees with the preceding interpretation based on the canonical coefficients of the $z_i^{(2)}$'s. The sample correlation between the two indices $\hat{U}_1$ and $\hat{V}_1$ is $\widehat{\rho_1^*} = .55$. There appears to be some overlap between job characteristics and job satisfaction.

- Scatter plots of the first $\left(\hat{U}_1, \hat{V}_1\right)$ pair may reveal atypical observations $\mathbf{x}_j$ requiring further study.
- If the canonical correlations $\widehat{\rho_2^*}, \widehat{\rho_3^*}, \ldots$ are also moderately large, scatter plots of the pairs $\left(\hat{U}_2, \hat{V}_2\right), \left(\hat{U}_3, \hat{V}_3\right), \ldots$ may also be helpful in this respect.
- Many analysts suggest plotting "significant" canonical variates against their component variables as an aid in subject-matter interpretation.

- If the sample size is large, it is often desirable to split the sample in halfs: the first half of the sample can be used to construct and evaluate the sample canonical variates and canonical correlations.
- The results can then be "validated" the remaining observations. The change (if any) in the nature of the canons analysis will provide an indication of the sampling variability and the stabilize.